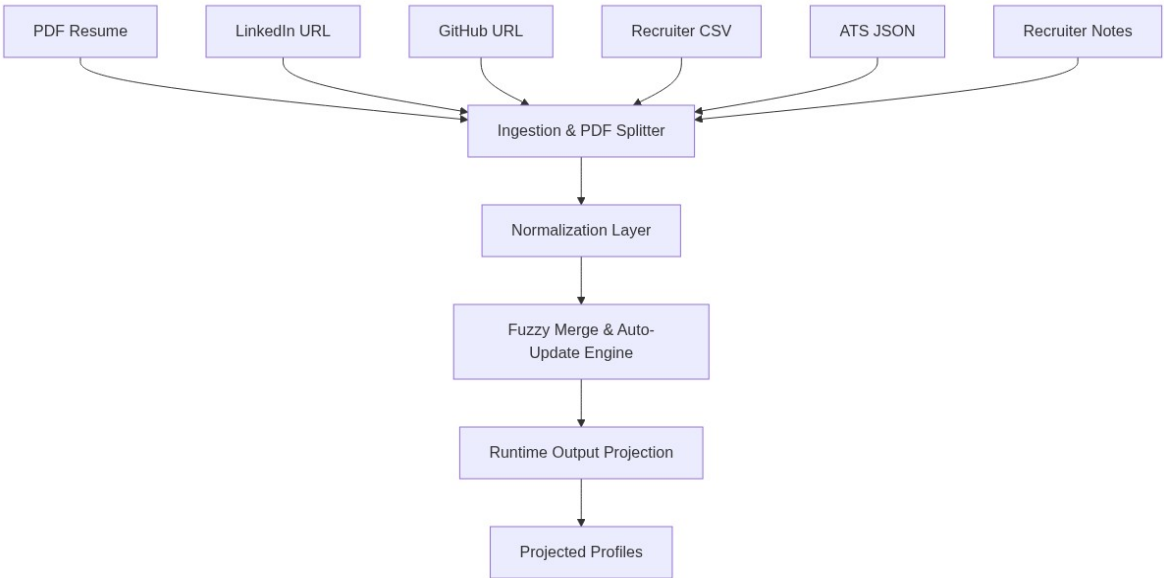


Technical Design: Candidate Profile Transformer

Candidate Name: Mathivanan S | Email: smadhivanan1234@gmail.com | Repo: candidate-transformer

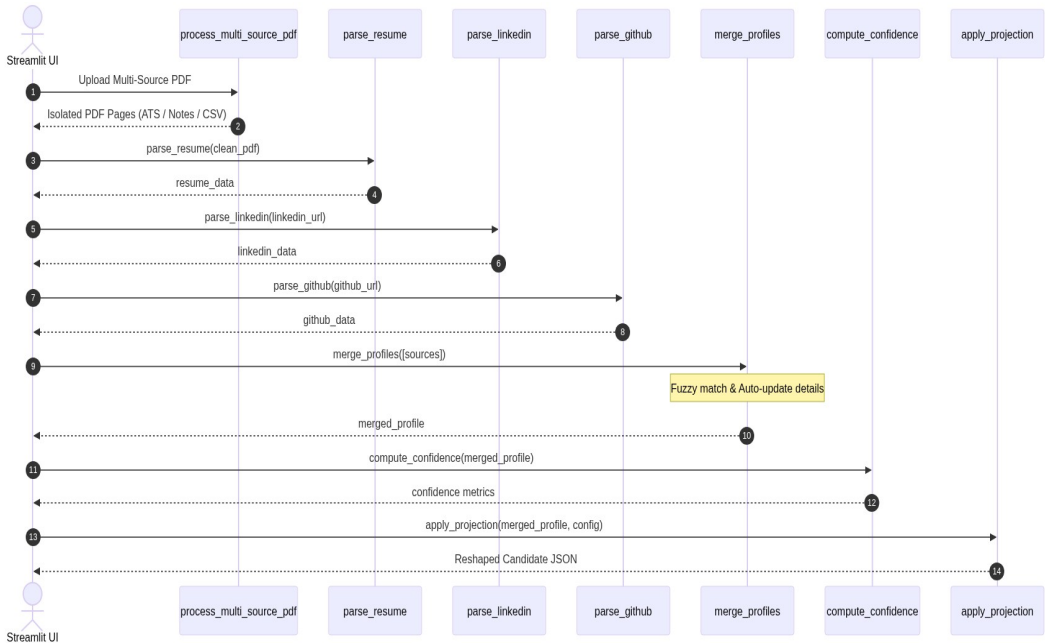
1. High-Level Design (HLD) Diagram

Conceptual block diagram representing source ingestion, split parsing, normalizations, and projection maps:



2. Low-Level Design (LLD) Sequence Flow

Sequential execution flow across parser modules, merge methods, and projection helpers:



3. Auto-Update & Field Enrichment Policy

- **Field Enrichment:** Missing parameters in low-priority profiles (e.g. ``linkedin_url``, ``github_url``) are automatically mapped and populated if found in other files.
- **Fuzzy Job Updates:** Company titles are cleaned (removing 'Systems', 'Inc') and matched by keyword intersections (e.g. 'Intern'). Amazon dates ('Present' → '2026-06-26') and detailed bullet descriptions auto-update from LinkedIn.
- **Project Linking:** Project titles are matched ignoring spacing and punctuation (e.g., matching 'FIND In' with 'FIND-In'). The GitHub URL and README description are automatically appended.
- **Lineage Logging:** Every modification adds a trace entry in the ``provenance`` metadata with a 'merge_update' method.

4. Canonical Output Schema & Normalizations

- **candidate_id:** Secure SHA-256 hash generated from standardized name and email.
- **phones:** Normalized to E.164 standards (e.g. ``+919087494162``).
- **location:** Standardized city and country codes mapped to ISO-3166 alpha-2 format.
- **skills:** Deduplicated list of skills. Bullet/heading keywords ('Languages', 'Frameworks') are filtered out.

5. Conflict Resolution & Confidence Policy

- **Priority Order:** Resume → LinkedIn → GitHub → CSV → ATS → Notes.
- **Scoring Formula:** Average score scaled by confirmations: $Confidence(v) = \min(1.0, AvgScore + 0.05 * (N - 1))$ where N is the confirming sources count.

6. Configurable Output (The Twist)

- **Shaping:** Supports index paths (e.g. ``emails[0]``) and flat list queries (e.g. ``skills[].name``).
- **Missing Policy:** Actions ``null`` (inserts null), ``omit`` (removes field), or ``error`` (aborts execution).

7. Scope Boundaries & Edge Cases

- **GitHub API Rate Limits:** Bypassed REST API limits (which block unauthenticated queries with HTTP 403) by implementing a BeautifulSoup HTML scraper to fetch profiles and repositories directly.
- **Multi-Source Ingestion:** Built an automated splitter that identifies ATS JSON, Recruiter Notes, or CSV pages within a single uploaded PDF. It extracts them into separate source profiles and excludes those pages from the main text stream, preventing parsing noise.
- **Fuzzy Job Merging (Amazon Intern vs SDE Intern):** Matches experience roles if company stems match and titles share keywords. Updates dates and descriptions from LinkedIn.
- **Headline Fallback (Contact Details Suppression):** If a resume header contains email, phone, or links on the second line, they are suppressed, and the parser falls back to using the Resume Summary as the headline.
- **Project Matching:** Consolidates project titles ignoring case, spaces, and punctuation (e.g., matching 'FIND In' with 'FIND-In') and merges README descriptions.

Deliberately Left Out (Under Time Constraint):

- **Real-time LLM parsing:** Omitted in favor of deterministic regex parsing to guarantee zero latency overhead, lower token costs, and 100% deterministic output.
- **Cross-Candidate Duplicate Resolution:** Focused strictly on single-candidate profile merges; database-wide de-duplication was moved out of scope.